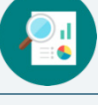
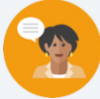


13.14 Using Coordinate Geometry to Solve Real-World Problems

Lesson Introduction

 Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.		 Learning Targets	- I can use coordinate geometry to classify triangles and quadrilaterals. - I can use coordinate geometry to solve real-world problems on the coordinate plane involving perimeter or area of polygons. - I can use coordinate geometry to solve real-world geometric problems involving circles, triangles, and quadrilaterals.
	MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.			
	MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.			
 Benchmark Coherence	Building On: MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.		Working Toward: N/A	
 Essential Questions	How can you use coordinate geometry to find the perimeter and area of composite figures in a real-world context?			
 Lesson Narrative	This is a cumulating lesson for unit 13. Students will be applying many of the skills and concepts to real-world problems.			
 Component Summary	13.14.1 Warm-Up (~5 mins)	Bell Work (<i>SE p. 360</i>)	13.14.3 Wrap-Up (~5 mins)	Reflection on Unit 13 (<i>SE p. 365</i>)
	13.14.2 Collaboration (40 mins)	Real-World Problems using Coordinate Geometry (<i>SE p. 361</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A		- Calculators (Optional) Graph paper or Graphing utility	
	<u>Additional Glossary:</u> N/A			
			Consider preparing small groups or partners for the Collaboration.	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may struggle with question 3 because not all the coordinate points are whole numbers. - Students may struggle with finding the distance of two points in order to find the area and/or perimeter. - Students may add the perimeter of each shape of a composite figure, instead of calculating the perimeter of the figure.		- Students may not recall secant, tangent, and inscribed angles for question 3 in 13.14.2 Collaboration. Encourage students to utilize their resources to find the information they need to complete the problem. - Consider providing graphing paper or a graphing utility to use as a visual for the beginning of question 2. - Consider working with students who are struggling during part of 13.14.2 Collaboration. - In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	MLR1: Stronger and Clearer Each Time Encourage students to revise and refine both their ideas and their verbal and written output for questions in 13.4.2 Collaboration. Suggest to students to first think or write individually about a response, then to refine and clarify the response through conversation with their partner, and then finally revise their original written response.		MA.K12.MTR.7.1: Real-World Contexts Students connect the mathematical concepts used in Unit 13 to solve real-world problems. Students use graphical models to understand, represent, and solve the problems. Teachers should monitor students for accuracy in both their models and their methods.	
 Differentiate Instruction	Collaboration 13.14.2 provides both visual and auditory entry points for learners to engage with the content. Engaging the class in discussions by having students explain their justifications and how they found the area and perimeter will help auditory learners deepen their understanding. The coordinate planes provide students with a visual to see each section of a composite figure. Consider placing the questions around the classroom to help kinesthetic learners be engaged in the lesson.			
	Additional differentiation opportunities are denoted throughout the lesson guide.			
Lesson Notes:				

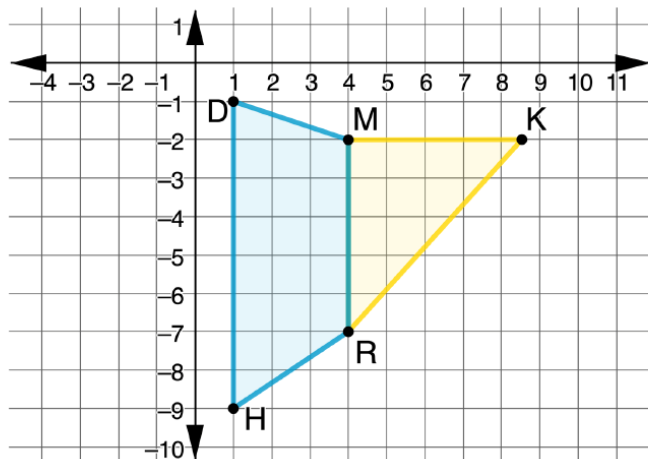
13.14.1 Warm-Up: Bell Work

Component Overview: Students are finding the area and perimeter of a composite figure that is created by a triangle and a trapezoid. Students should complete the questions independently.



13.14.1 Warm-Up: Bell Work

- The composite figure shown is created by $\triangle MKR$ and trapezoid $DMRH$. $M(4, -2)$, $K(8.5, -2)$, $R(4, -7)$, $H(1, -9)$, and $D(1, -1)$.



- Complete the table. If necessary, round to the nearest thousandth.

Figure	Perimeter	Area
$\triangle MKR$	16.227 units	11.25 square units
$DMRH$	19.768 units	19.5 square units

- What is the perimeter of the composite figure? If necessary, round to the nearest thousandth.
25.995 units
- What is the area of the composite figure? If necessary, round to the nearest thousandth.
30.75 square units



Use student understanding to inform today's Collaboration. Students who struggled on the Bell Work may benefit from working with you in a small group on part of the Collaboration. Make a list of misconceptions and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

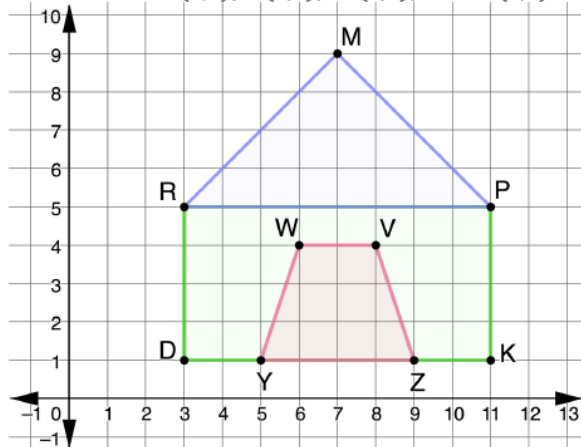
13.14.2 Collaboration: Real-World Problems

Component Overview: This Collaboration provides students with multiple opportunities to work with composite figures in the real world. Throughout each real-world prompt, students are asked to classify polygons, find the area and perimeter of polygons and composite figures, and find additional information while justifying their answers along the way.



13.14.2 Collaboration: Real-World Problems

1. Lana is designing the front of a doghouse. The computer software generated coordinate points where each unit equals a foot. The roof has vertices at $M(7,9)$, $P(11,5)$, and $R(3,5)$. The front wall has vertices at $P(11,5)$, $K(11,1)$, $D(3,1)$, and $R(3,5)$. The door entrance has vertices at $Z(9,1)$, $Y(5,1)$, $W(6,4)$, and $V(8,4)$.



- a. What type of triangle is the roof, $\triangle MPR$? Justify your answer.
Isosceles triangle because $MR = MP = \sqrt{32}$
- b. What is the area of $\triangle MPR$?
16 square feet
- c. Clay tiles cost \$8 per square foot. How much will it cost Lana to put clay tiles on the given face of the doghouse's roof?
\$128.00



Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions, and which students help clarify. Consider having an answer key available for students to verify their answers when they have completed all of the parts of a question. Consider requiring students to receive your signature before moving on to the next question. This provides some accountability, while also allowing you to ensure students are understanding. As you review student work, consider which parts students should revise before moving on to the next question.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Collaboration.

13.14.2 Collaboration: Real-World Problems (continued)

- d. What type of quadrilateral is the door, $WVZY$? Justify your answer.

An isosceles trapezoid because both $\overline{WV}$ and $\overline{YZ}$ have a slope of 0 so $\overline{WV} \parallel \overline{YZ}$ and $WY = VZ = \sqrt{10}$.

- e. Lana is placing a frame around the door, $WVZY$. How many feet of wood will she need? If necessary, round to the nearest thousandth.

12.325 feet

- f. What type of quadrilateral is the front wall, $RPKD$. Justify your answer.

$RPKD$ is a rectangle because $RD = PK = 4$ and $RP = DK = 8$, and the diagonals are congruent, $DP = RK = 4\sqrt{5}$.

- g. Lana is painting the front wall, $RPKD$, excluding the door, $WVZY$. What is the area of the front wall Lana painting?

23 square feet

- h. Paint costs \$0.55 per square foot. Excluding the door, how much will it cost to paint the front wall?

\$12.65

2. Justin is creating a layout for his garden in the shape of a quadrilateral, $GWDP$, with vertices $G(1,8)$, $W(11,8)$, $D(11,0)$, and $P(1,0)$.

- a. Classify $GWDP$. Justify your answer.

Quadrilateral $GWDP$ is a rectangle because $GW = DP = 10$ and $WD = GP = 8$, and the diagonals are congruent, $DG = PW = 2\sqrt{41}$.



Consider challenging students by adding an extension to question 1h:

- Lana found another company that sells paint for \$0.05 per square inch. If Lana is covering 23 square feet, is this a better deal than \$0.55 per square foot? Justify your answer. (Answer: \$0.05 per square inch is \$0.60 per square foot, so it is not a better deal.)



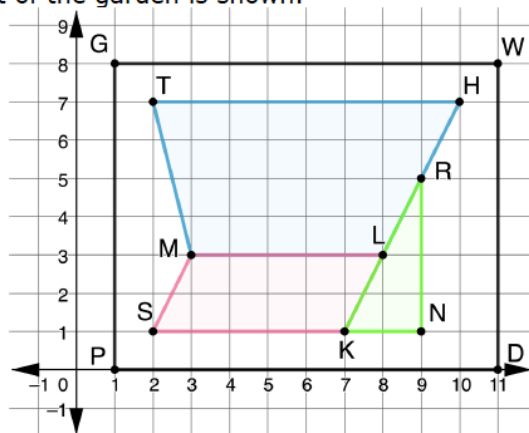
Struggling students may benefit from using graphing paper or a graphing utility for question 2. This will provide students with a visual and could help them to see the three sections of the composite figure.

13.14.2 Collaboration: Real-World Problems (continued)

Justin created three sections in the garden: $THLM$, $LKSM$, and $\triangle KNR$.

- b. $THLM$ has vertices at $T(2,7)$, $H(10,7)$, $L(8,3)$, and $M(3,3)$. Classify $THLM$. Justify your answer.
Quadrilateral $THLM$ is a Trapezoid because both $\overline{TH}$ and $\overline{ML}$ have a slope of 0 so $\overline{ML} \parallel \overline{TH}$.
- c. $LKSM$ has vertices at $L(8,3)$, $K(7,1)$, $S(2,1)$, and $M(3,3)$. Classify $LKSM$. Justify your answer.
Quadrilateral $LKSM$ is a parallelogram because both $\overline{MS}$ and $\overline{LK}$ have a slope of 2 so $\overline{MS} \parallel \overline{LK}$ and both $\overline{ML}$ and $\overline{SK}$ have a slope of 0 so $\overline{ML} \parallel \overline{SK}$.
- d. $\triangle KNR$ has vertices at $K(7,1)$, $N(9,1)$, and $R(9,5)$. Classify $\triangle KNR$. Justify your answer.
Triangle KNR is a right triangle. The slope of $\overline{KN}$ is 0 and the slope of $\overline{RN}$ is undefined, therefore the two lines are perpendicular and form a right angle.

The layout of the garden is shown.



- e. If one square on the grid represents one square foot, how many feet of fencing will Justin need to go around $GWDP$?
36 feet



Students may struggle with understanding “one square on the grid represents one square foot.” Listen for students who are confused by this, and take the time to show them that each square on the grid is 1 unit by 1 unit, so the area is 1 square unit.

13.14.2 Collaboration: Real-World Problems (continued)

- f. Complete the table. If necessary, round to the nearest thousandth.

Section	Perimeter	Area
THLM	21.595 feet	26 square feet
LKSM	14.472 feet	10 square feet
ΔKNR	10.472 feet	4 square feet

- g. What is the perimeter of the composite figure made by the three sections of the garden? If necessary, round to the nearest thousandth.

27.595 feet

- h. What is the area of the three sections of the garden? If necessary, round to the nearest thousandth.

40 square feet

- i. If a 40 pound (lb) bag of soil covers 12 square feet, how many bags of soil will Justin need to cover the three sections of the garden?

He would need 4 bags because he needs exactly $3\frac{1}{3}$ bags so he would round up since the bags must be bought in whole numbers.



Consider having students use the instructional routine, “Take Turns,” on question 2f. Each person in the group will take a turn at finding the perimeter and/or the area. Each group member can use this time to review each other’s work and ask clarifying questions, if needed.

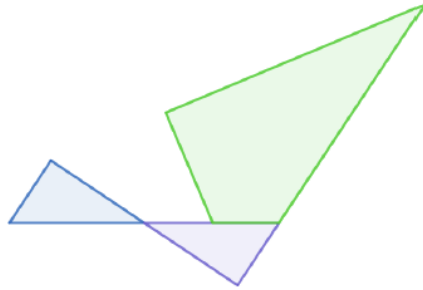


Consider prompting questions to help students make sense of the problem and as a check for understanding:

- How is finding the perimeter of a composite figure different than finding the area of a composite figure?
- When finding the number of bags of soil, would you use the area or perimeter? Why?
- Why would you round up for the number of bags?

13.14.2 Collaboration: Real-World Problems (continued)

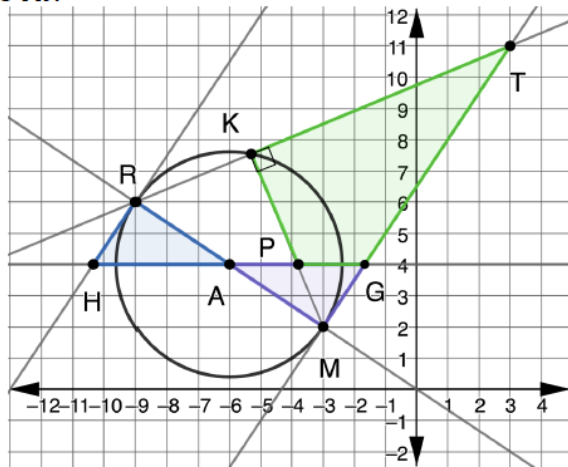
3. Edwin started creating a new logo design for the company office using the images shown.



To determine how to place the logo on the wall in the company's office, he uploaded his design to a computer program and got the following coordinates.

$M(-3,2)$	$R(-9,6)$	$P(-3.8,4)$	$T(3,11)$
$G(-\frac{5}{3},4)$	$H(-\frac{31}{3},4)$	$K(-5.3,7.5)$	$A(-6,4)$

Circle A has points R , M , and K on the circle. Lines TM and HR are tangent to circle A . Lines RM and HG are secant to circle A . Line segment KM is perpendicular to line TK .



Students may struggle with this question because some of the points are not whole numbers and cannot be easily identified on the coordinate plane. If necessary, support these students by incorporating the Desmos scientific calculator, which can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.



Consider asking guided questions to help students make sense of the information given:

- What relationship does a tangent line have with the radius/diameter of a circle?
- What relationship does $\angle RKM$ have with the diameter of the circle?
- How can the diameter be used to find the $m\angle MKT$?

13.14.2 Collaboration: Real-World Problems (continued)

- a. Complete the table by finding the area of each triangle.

Triangle	Work	Area
$\triangle AHR$	Answers will vary. $\text{Area } \triangle AHR = \frac{1}{2} \cdot HA \cdot 2$	≈ 4.333 square units
$\triangle AMG$	Answers will vary. $\text{Area } \triangle AMG = \frac{1}{2} \cdot AG \cdot 2$	≈ 4.333 square units
$\triangle PGM$	Answers will vary. $\text{Area } \triangle PGM = \frac{1}{2} \cdot PG \cdot 2$	≈ 2.133 square units
$\triangle TKM$	Answers will vary, depending on rounding. $\text{Area } \triangle TKM = \frac{1}{2} \cdot TK \cdot MK$ $TK \approx 9.008$ $MK \approx 5.961$	≈ 26.85 square units



Consider having students use the instructional routine, “Take Turns,” on question 3a. Each person in the group will take a turn at finding the area of each triangle. Each group member can use this time to review each other’s work and ask clarifying questions, if needed.

- b. Explain how to use $\triangle PGM$ and $\triangle TKM$ to find the area of $TKPG$.

Take the area of triangle TKM and subtract the area of triangle PGM to find the area of polygon $TKPG$

- c. What is the area of $TKPG$? If necessary, round to the nearest thousandth.

$$26.85 - 2.133 \approx 24.717 \text{ square units}$$

- d. What is the total area of Edwin’s design? If necessary, round to the nearest thousandth.

$$24.717 + 4.333 + 4.333 \approx 33.383 \text{ square units}$$

13.14.3 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment based on their level of understanding of each key concept from Unit 13. It is designed to be done independently to gather specific, student-level data.



13.14.3 Wrap-Up: Reflection

Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Using coordinate geometry to determine if lines are parallel or perpendicular.	  
Using coordinate geometry to partition a line segment or find the weighted average.	  
Using coordinate geometry to classify triangles or quadrilaterals.	  
Using coordinate geometry to justify definitions, properties, and theorems involving circles, triangles or quadrilaterals.	  
Using coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	  
Using coordinate geometry to solve mathematical and real-world involving perimeter or area of polygons.	  

Answers will vary.



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.